

PART A - Circuit calculations

d) Series circuit : 125 V source with 20 Ω , 30 Ω 50 Ω in series

1 Total resistance

$$R_T = R_1 + R_2 + R_3 = 20 + 30 + 50 = 100 \Omega$$

2 Total current from the supply

$$I_T = \frac{V}{R_T} = \frac{125}{100} = 1.25 \text{ A}$$

3 Current through each resistor

In a circuit, current is the same through all components

$$I_{20\Omega} = I_{30\Omega} = I_{50\Omega} = 1.25 \text{ A}$$

4 Voltage drop through each resistor

$$V = IR$$

$$20 \Omega = 1.25 \times 20 = 25 \text{ V}$$

$$30 \Omega = 1.25 \times 30 = 37.5 \text{ V}$$

$$50 \Omega = 1.25 \times 50 = 62.5 \text{ V}$$

$$25 + 37.5 + 62.5 = 125 \text{ V}$$

5 Power dissipated through each resistor

$$P = V \times I$$

$$20 \Omega = 25 \times 1.25 = 31.25 \text{ W}$$

$$30 \Omega = 37.5 \times 1.25 = 46.875 \text{ W}$$

$$50 \Omega = 62.5 \times 1.25 = 78.125 \text{ W}$$

$$P_T = 31.25 + 46.875 + 78.125 = 156.25 \text{ W}$$

$$P_T = V I_T = 1.25 \times 125 = 156.25 \text{ W}$$

b) Parallel circuit

1) Total resistance

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$\frac{1}{R_T} = \frac{5}{100} + \frac{1}{100} + \frac{2}{100} = \frac{8}{100}$$

$$R_T = \frac{100}{8} = 12.5 \Omega$$

$$\frac{1}{0.08} = 12.5 \Omega$$

2) Total current from the supply

$$I_T = \frac{V}{R_T} = \frac{125}{12.5} = 10 \text{ A}$$

3) Current through each resistor

$$I = \frac{V}{R}$$

$$20 \Omega = \frac{125}{20} = 6.25 \text{ A}$$

$$100 \Omega = \frac{125}{100} = 1.25 \text{ A}$$

$$50 \Omega = \frac{125}{50} = 2.5 \text{ A}$$

4 Voltage drop through each resistor $25 \times 5 = 125$

All parallel branches have same voltage

$$V_{20\Omega} = V_{100\Omega} = V_{50\Omega} = 125 \text{ V}$$

5 Power dissipated through each resistor

$$P = VI$$

$$20 \Omega = 125 \times 0.25$$

$$781.25 \text{ W}$$

$$100 \Omega = 125 \times 1.25 = 156.25 \text{ W}$$

$$50 \Omega = 125 \times 2.5 = 312.5 \text{ W}$$

$$P_T = 781.25 + 156.25 + 312.5 = 1250 \text{ W}$$